

**Sixth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

OBJECT ORIENTED DATA STRUCTURE

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated against them.
 - (2) Assume suitable data and illustrate answers with Java code and diagrams wherever necessary.
-
1.
 - (a) What is Object Oriented Programming ? Differentiate between Procedure Oriented Programming and Object Oriented Programming based on six different parameters. 3(CO1)
 - (b) Explain access modifiers used in Java. Draw the table showing the accessibility inside and outside the package. 3(CO1,2,)
 - (c) An election is contested by 5 candidates. The candidates are numbered as 1 to 5 and the voting is done by marking the candidate number on the ballot paper. Write a program to read the ballots and count the votes cast for each candidate using an array variable count. In case, if a number read is outside the range from 1 to 5, then the ballot should be considered as a “spoilt ballot” and also the program should count the number of spoilt ballot. 4(CO2)
 2.
 - (a) Create a class **Movie** with data members as movie_number, movie_name, release_year, and budget. Include methods to input information for N movies and to print the information of movies having a budget between 3,00,00,000 to Rs. 5,00,00,000. Write a complete Java program to test the class.(Use an array of objects). 5(CO2)
 - (b) Create a class named as **Pizza** that stores details about a pizza. It should contain the following:
Instance Variables :
String pizza_size_to store the size of the pizza(small, medium or large)
int cheese – the number of cheese toppings

int pepperoni - the number of pepperoni toppings

int mushroom - the number of mushroom toppings

Member Methods :

Constructor - to initialize all the instance variables

CalculateCost() - A public method that returns a double value, that is, the cost of the pizza.

The Pizza cost is calculated as follows:

- Small: Rs500 + Rs.25 per topping
- Medium : Rs.650 + Rs.25 per topping
- Large: Rs.800 + Rs.25 per topping

PizzaDescription() – A public method that returns a String containing the pizza size, quantity of each topping, and the pizza cost as calculated by CalculateCost().

Write a complete Java program for the above scenario. 5(CO2)

3. (a) Illustrate the use of a default(non-parametized) constructor in Multilevel Inheritance with proper example. Which constructor is used first and why? 2(CO2)

- (b) Define an Interface **Queue** having methods:

(i) void insert(int item)

(ii) int delete()

Create a class called as QueueDemo that implements the above interface to implement queue operations. 4(CO2)

- (c) Consider the three classes Employee, Salaried_Employee, and Daily Wages_Employee. Find out the relationship among the classes and apply the appropriate object-oriented concept to write a program that calculates and prints the salary of 100 employees. [Assume suitable methods and instance variables]. 4(CO2)

4. (a) What is the need for asymptotic notations in the algorithm ? Describe commonly used asymptotic notations with their significance. 5(CO3)

(b) Find the time complexity for the given piece of code and elaborate it:

(i) class GFG

```
{  
    Public static void main(Sting[]args)  
    {  
        int i, n = 8;  
        for(i = 1,i <= n; i++){  
            System. out.printf(“Hello World!!!\n”);  
        }  
    }  
}
```

(ii) import java.lang.Math;

```
class GFG  
{  
    Public static void main(String args[])  
    {  
        in i, n=8;  
        for(i = 2; i = (int)Math.pow(i,2){  
            System.out println(“Java Programming!!!”);  
        }  
    }  
}
```

5(CO3)

5. (a) Write a Java method or algorithm to implement primitive operations of the stack.

Convert the following Infix Expression to its Postfix Expression using stack.[Show all the steps of conversion using stack]

(A + B) * C –(D – E) \$ (F + G)

5(CO4)

- (b) Compare Double-Ended Queue and Circular Queue with proper diagram. Also, write an algorithm to insert and delete elements using Circular Queue. 5(CO4)
6. (a) Consider a Singly Linked Linear List. Write Java method or algorithm:
- (i) To create a a list using element insertion at the end.
 - (ii) To search for an element from the list. 5(CO4)
- (b) Write an algorithm for the deletion of the middle node in the Doubly Linked List. 5(CO4)

